

FORT DODGE, IA, USA

WMO: 725490

Lat: 42.550N

Lon: 94.183W

Elev: 1156

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.0	(c) -5.1	(d) -17.6	(e) 2.2	(f) -8.2	(g) -12.2	(h) 3.0	(i) -4.6	(j) 24.7	(k) 25.1	(l) 22.4	(m) 25.5	(n) 5.0	(o) 320	(p) 0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	90.8	75.5	88.4	74.3	85.6	73.0	78.3	87.2	76.5	85.4	74.8	83.1	12.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	138.5	84.3	73.2	129.0	82.1	72.2	124.3	81.0	42.9	88.2	40.8	85.5	39.0	82.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.3	(b) 19.9	(c) 18.0	(e) -17.1	(f) 95.3	(g) 5.2	(h) 3.2	(i) -20.9	(j) 97.6	(k) -23.9	(l) 99.5	(m) -26.9	(n) 101.2	(o) -30.6	(p) 103.6		
(4) 23.3	(5) 19.9	(6) 18.0	(7) -17.7	(8) 81.6	(9) 4.9	(10) 1.7	(11) -21.3	(12) 82.9	(13) -24.1	(14) 83.9	(15) -26.9	(16) 84.9	(17) -30.5	(18) 86.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.9	18.2	22.4	35.4	48.1	59.8	69.8	73.5	70.8	63.9	50.8	36.6	23.5	
	DBStd	21.71	12.92	12.85	12.30	10.27	8.54	6.49	5.85	5.50	8.46	9.73	10.83	11.89	
	HDD50	3758	986	773	477	153	15	0	0	0	5	110	419	820	
	HDD65	7042	1451	1193	919	515	209	27	5	12	121	453	852	1285	
	CDD50	2977	0	0	24	96	318	594	729	645	421	133	17	0	
	CDD65	784	0	0	1	7	46	171	269	192	87	11	0	0	
	CDH74	7315	0	0	9	116	485	1610	2535	1580	852	125	3	0	
	CDH80	2390	0	0	0	28	132	540	921	478	260	31	0	0	
Wind	WSAvg	8.1	8.4	8.6	9.3	10.4	9.1	7.7	6.3	5.8	6.9	8.3	8.7	8.2	
Precipitation	PrecAvg	33.7	0.9	0.9	2.1	3.3	4.4	5.0	4.5	4.3	3.3	2.2	1.7	1.1	
	PrecMax	48.2	2.6	3.1	6.1	8.5	10.3	12.0	11.3	11.4	13.7	6.1	5.3	2.9	
	PrecMin	21.9	0.0	0.2	0.0	0.6	1.7	1.8	1.1	0.5	0.4	0.2	0.0	0.0	
	PrecStd	6.5	0.6	0.7	1.4	1.8	2.0	2.5	2.2	2.3	2.6	1.5	1.3	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	57.4	73.5	83.7	90.0	92.7	95.4	92.6	90.7	84.0	71.7	56.5	
		MCWB	42.5	49.5	58.7	63.9	70.4	75.1	78.9	76.4	72.0	66.1	58.4	51.0	
	2%	DB	43.4	50.0	65.9	76.9	83.6	89.7	91.1	89.1	87.5	77.1	63.6	48.0	
		MCWB	38.1	45.0	55.4	59.8	67.0	73.1	77.6	76.5	72.1	62.1	54.1	43.1	
	5%	DB	37.5	44.7	60.6	71.8	79.5	86.0	88.5	84.4	82.2	72.2	57.2	43.5	
		MCWB	34.1	39.8	52.4	56.7	63.8	71.6	76.3	74.2	68.7	60.5	48.9	38.7	
	10%	DB	35.6	38.8	53.9	65.9	75.1	82.2	84.5	81.9	79.2	66.3	53.6	38.7	
		MCWB	33.2	35.4	47.2	54.0	61.6	69.1	74.1	72.0	66.8	57.0	46.8	35.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	51.9	62.7	66.1	73.3	78.5	82.1	80.3	75.7	70.4	59.8	52.5
			MCDB	50.2	56.3	70.2	80.2	85.4	88.4	91.0	88.3	86.1	79.7	67.8	55.7
2%		WB	38.6	45.0	57.4	61.7	69.4	75.7	79.3	77.6	73.5	65.5	54.8	44.3	
		MCDB	43.0	49.7	64.8	72.9	81.0	85.3	87.9	86.4	83.0	73.9	61.9	47.6	
5%		WB	35.1	39.6	52.5	58.5	66.9	73.5	77.5	75.3	71.3	61.7	50.3	39.1	
		MCDB	37.9	44.1	60.0	69.3	76.3	83.2	86.5	83.1	79.7	69.4	56.8	43.4	
10%		WB	32.4	35.6	46.7	54.6	63.9	71.3	75.5	73.6	68.8	58.2	46.2	35.5	
		MCDB	34.7	39.0	53.0	64.4	72.3	80.5	84.0	80.5	76.8	66.2	51.9	39.0	
Mean Daily Temperature Range	5% DB	MDBR	16.2	16.8	18.1	21.5	20.6	20.1	19.5	19.3	22.4	21.0	18.1	15.4	
		MCDBR	20.3	21.1	27.4	30.9	25.8	23.1	21.4	21.7	25.0	27.2	26.1	21.2	
		MCWBR	16.6	16.9	18.1	17.7	13.4	11.4	11.4	11.9	12.8	15.7	17.4	17.0	
	5% WB	MCDBR	19.4	20.2	25.6	25.9	21.8	20.0	19.4	19.4	21.2	22.6	23.3	20.2	
		MCWBR	16.2	16.4	17.8	16.7	13.0	11.8	11.4	11.6	12.1	15.1	17.0	16.6	
Clear-Sky Solar Irradiance	taub	0.293	0.288	0.317	0.369	0.396	0.418	0.426	0.413	0.383	0.339	0.296	0.277		
	taud	2.383	2.418	2.438	2.321	2.313	2.297	2.249	2.285	2.328	2.446	2.507	2.460		
	Ebn at Noon	269	292	295	285	279	272	268	268	268	269	268	267		
	Edh at Noon	25	29	32	39	40	41	43	40	36	28	22	21		
All-Sky Solar Radiation	RadAvg	596	924	1201	1506	1778	1970	2036	1730	1413	949	641	481		
	RadStd	61	73	120	136	130	160	116	119	96	121	46	36		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page